

MOSAIC: Multi-Modal Open Surveillance with AI-Driven Calibrated Inference

A Calibrated Bayesian Fusion of Wastewater, Genomic, and Text Streams for Outbreak Early Warning

Aravind Kannappan

MOSAIC Project • <https://github.com/aravinds-kannappan/MOSAIC>

Abstract

Pandemic early-warning systems fail less often from a lack of data than from a lack of *integration* and *calibration*: heterogeneous signals—wastewater concentrations, pathogen genomes, and outbreak text reports—arrive on different cadences, in different units, and with different noise models, and the probabilities ultimately presented to decision-makers are rarely validated. We present MOSAIC, an open-source system that fuses three public, authentication-free surveillance streams into a single calibrated posterior, the probability that the effective reproduction number exceeds one, $\mathbb{P}(R_t > 1)$. We give the probabilistic model in full: a renewal-equation latent-incidence process with stream-specific observation kernels (negative-binomial for wastewater, Poisson for text-event counts, Dirichlet–multinomial for lineage frequencies), Bayesian online change-point detection (BOCPD) on the wastewater and text streams, a Jensen–Shannon divergence anomaly detector on the genomic stream, an EpiEstim reproduction-number estimator, and an independent-evidence fusion rule that reduces, in the full backend, to a hierarchical model fit by the No-U-Turn Sampler. We then evaluate the deployed system on real data. Calibrating $\mathbb{P}(R_t > 1)$ as a probabilistic forecast of realised growth on the multi-year CDC National Wastewater Surveillance System (NWSS) record ($N=1,334$ day-ahead forecasts) yields an Expected Calibration Error of 0.086, a Brier score of 0.124, and an AUROC of 0.917—the stated probabilities are well-calibrated and strongly discriminative. On the live feed the system surfaces concurrent real outbreaks (Bundibugyo ebolavirus in the Democratic Republic of the Congo, hantavirus clusters, an elevated U.S. SARS-CoV-2 wastewater wave) with per-stream attribution. We also report a corrected numerical issue in the reproduction-number tail probability that we recommend other lightweight implementations check. All figures in this paper are computed from real public data; nothing is synthetic.

1 Introduction

Every major epidemic of the past two decades was preceded by signals that were, in retrospect, detectable days to weeks before official declarations. The binding constraint on early warning is increasingly not the existence of data: municipal wastewater is sampled routinely [13], pathogen genomes are deposited continuously and shared openly [8], and structured outbreak reports are published by the World Health Organization and by moderated mailing lists [15]. The constraint is *fusion*—turning three asynchronous, differently-scaled, differently-noised streams into one coherent quantity—and *calibration*—ensuring that when the system states “78%,” the event happens about 78% of the time.

We argue that the right target quantity for a decision-maker is the posterior probability that transmission is currently growing,

$$p_t = \mathbb{P}(R_t > 1 \mid \text{data up to } t), \quad (1)$$

because (i) it is unitless and comparable across pathogens and locations, (ii) it has an operational meaning—the probability the epidemic curve is turning upward—and (iii) it is directly falsifiable against subsequently observed growth, which makes *calibration* measurable rather than aspirational.

Contributions.

1. A complete probabilistic specification (§4) for multi-modal outbreak inference: a renewal latent-incidence process with negative-binomial (wastewater), Poisson (text), and Dirichlet–multinomial (genomic) observation kernels, plus BOCPD, Jensen–Shannon, and EpiEstim detectors, and a fusion rule that reduces to a hierarchical model under the full backend.
2. A two-tier open implementation (§5): a serverless TypeScript deployment that runs the entire pipeline in-browser/edge, and a Python backend that performs the full No-U-Turn-Sampler posterior, with the former designed to be mathematically faithful to the latter.
3. An empirical calibration study (§6) on the real multi-year NWSS record showing $\mathbb{P}(R_t > 1)$ is well-calibrated (ECE = 0.086) and discriminative (AUROC = 0.917), together with live multi-stream alert attribution and a documented numerical correction.

2 Related Work

Wastewater-based epidemiology. Wastewater concentration tracks community prevalence with a lead over clinical case reports and is now a standing U.S. surveillance program [12, 13]. Most public tools present concentrations or qualitative trends; we instead convert the national percentile record into a calibrated growth probability.

Reproduction-number estimation. The instantaneous reproduction number R_t is most commonly estimated by the EpiEstim sliding-window Poisson–Gamma method of Cori et al. [4], building on renewal-equation theory [16]. We use EpiEstim as the renewal core and treat $\mathbb{P}(R_t > 1)$ as the primary output.

Change-point and anomaly detection. Bayesian Online Change-Point Detection [1] provides a recursive run-length posterior under conjugate emissions; we apply it with Poisson–Gamma emissions to event-count and wastewater series. Distributional shift in categorical lineage frequencies is naturally measured by the Jensen–Shannon divergence [11], which is bounded and symmetric.

Probabilistic forecast calibration. The reliability diagram, the Brier score [3], proper scoring rules [6], and the Expected Calibration Error [7] are the standard instruments for asking whether stated probabilities match observed frequencies; modern neural systems are frequently miscalibrated [7], which motivates measuring ECE explicitly here.

Probabilistic programming. The full hierarchical posterior is sampled with the No-U-Turn Sampler [10] as implemented in NumPyro [2, 14].

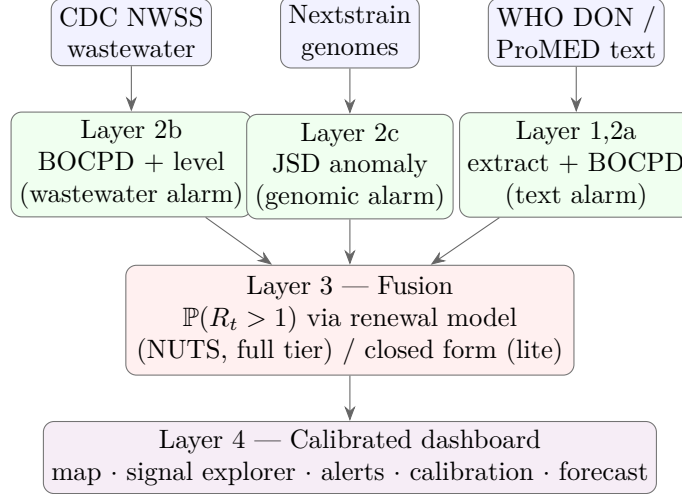


Figure 1: MOSAIC architecture. Three public streams feed three modality-specific detectors (Layer 2), whose alarms are fused into the growth posterior $\mathbb{P}(R_t > 1)$ (Layer 3) and surfaced by the dashboard (Layer 4). The lightweight serverless tier and the full NumPyro tier share observation models and target quantity.

3 System Architecture

MOSAIC is organized as four layers (Figure 1). Layer 1 extracts structured outbreak events from text. Layer 2 runs three independent detectors, one per modality. Layer 3 fuses the detector outputs into the posterior of Eq. (1). Layer 4 is a calibrated dashboard.

A central design decision is that the system runs in two faithful tiers. A *lightweight tier* executes the full pipeline in TypeScript on serverless infrastructure with no backend, so the public dashboard is always live; a *full tier* runs the hierarchical NumPyro posterior in Python when a backend URL is configured. The two tiers share the same observation models and the same target quantity; the lightweight tier replaces the joint sampler with conjugate/closed-form approximations (§4.6). All data sources are public and require no authentication.

4 Probabilistic Model and Methods

4.1 Latent incidence and the renewal equation

Let $I_t \geq 0$ denote latent new infections on day t in a given location–pathogen cell. The renewal equation links incidence to the instantaneous reproduction number R_t through the generation-interval (serial interval) distribution $w_{1:S}$ [4, 16]:

$$\mathbb{E}[I_t \mid R_t, I_{<t}] = R_t \sum_{s=1}^S w_s I_{t-s}, \quad \log R_t \sim \mathcal{N}(\log R_{t-1}, \sigma_R^2), \quad (2)$$

where $w_s \geq 0$, $\sum_s w_s = 1$, is obtained by discretising a Gamma serial interval (for SARS-CoV-2, mean 5.1, sd 4.0 days [9]) via midpoint quadrature, $w_s \propto f_{\text{Gamma}}(s; \kappa, \theta)$. The total infectiousness at t is $\Lambda_t = \sum_{s \geq 1} w_s I_{t-s}$. The three streams are conditionally independent observations of the same latent $I_{1:T}$.

4.2 Wastewater stream: NWSS aggregation, BOCPD, and sustained elevation

National aggregation. The CDC NWSS dataset reports, per site and per day, a *percentile* $c_{i,t} \in [0, 100]$ giving the current 15-day-averaged viral activity relative to that site’s own history. We form a national signal by server-side aggregation,

$$\bar{c}_t = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} c_{i,t}, \quad (3)$$

over all sites $\mathcal{S}_t$ reporting on day t (typically $|\mathcal{S}_t| \approx 1,200$). Aggregating server-side keeps the payload small and yields a dense daily series spanning multiple waves (Figure 2).

Observation kernel. Conditioned on latent incidence, wastewater concentration is over-dispersed count-like and lagged; we model

$$C_t \sim \text{NegBin}(\rho_W I_{t-d_W}, \phi_W), \quad (4)$$

with shedding scale ρ_W , reporting delay d_W , and dispersion ϕ_W .

Bayesian online change-point detection. To flag abrupt onsets we run BOCPD [1] on the discretised series $n_t = \lfloor \bar{c}_t \rfloor$. With a Poisson likelihood $n_t | \lambda \sim \text{Poisson}(\lambda)$ and a conjugate $\lambda \sim \text{Gamma}(\alpha_0, \beta_0)$ prior, the run-length posterior $P(r_t | n_{1:t})$ is propagated recursively. Writing r_t for the time since the last change-point and using a constant hazard $H = 1/\mu_{\text{RL}}$,

$$P(r_t=0, n_{1:t}) = \sum_{r_{t-1}} P(r_{t-1}, n_{1:t-1}) H \pi(n_t | r_{t-1}), \quad (5)$$

$$P(r_t=r_{t-1}+1, n_{1:t}) = P(r_{t-1}, n_{1:t-1}) (1-H) \pi(n_t | r_{t-1}), \quad (6)$$

where the posterior predictive of the Poisson–Gamma model is the negative-binomial

$$\pi(n | \alpha, \beta) = \text{NegBin}\left(n; \alpha, \frac{\beta}{\beta+1}\right) = \frac{\Gamma(\alpha+n)}{\Gamma(\alpha) n!} \left(\frac{\beta}{\beta+1}\right)^\alpha \left(\frac{1}{\beta+1}\right)^n, \quad (7)$$

and sufficient statistics update as $\alpha \leftarrow \alpha + n$, $\beta \leftarrow \beta + 1$ along each run. The instantaneous change-point probability is $P(r_t=0 | n_{1:t})$; the stream *alarm* is the maximum of this quantity over a recent window of length W , $a_t^{\text{ww,cp}} = \max_{t-W < \tau \leq t} P(r_\tau=0)$. Using the maximum (rather than a cumulative product) is deliberate: a cumulative combination of per-step probabilities drifts toward one even under the null, whereas the windowed maximum isolates genuine recent shifts.

Sustained elevation. BOCPD fires on *abrupt* change but a slowly-rising, persistently-elevated wave is also informative. Because the percentile $\bar{c}_t$ is already a site-relative level, we add an elevation alarm $a_t^{\text{ww,lv1}} = \text{clip}((\bar{c}_t - 50)/40, 0, 1)$ and combine the two by the noisy-or rule $a_t^{\text{ww}} = 1 - (1 - a_t^{\text{ww,cp}})(1 - a_t^{\text{ww,lv1}})$.

4.3 Text stream: extraction, change-points, and recency weighting

Extraction (Layer 1). Outbreak reports from the WHO Disease Outbreak News API (ordered most-recent-first) and the ProMED posts API are parsed into structured events (pathogen, location, date, counts, no). The full backend uses a constrained LLM extractor; the lightweight tier uses curated regular expressions plus a country resolver that maps free text to ISO-3166 codes. Daily per-pathogen event counts $e_{k,t}$ are the detector input.

Observation kernel and change-points. Reporting intensity is modelled as Poisson in latent incidence,

$$E_t \sim \text{Poisson}(\lambda_N \hat{q}_t I_{t-d_N}), \quad (8)$$

with reporting rate λ_N , ascertainment $\hat{q}_t$, and delay d_N . We run the same Poisson–Gamma BOCPD of Eq. (6) on the dense, zero-filled daily count series, which makes a burst of reports after a quiet period a strong change-point (Figure 5).

Recency/intensity weighting. Because outbreak reports are sparse and sporadic, the operational text alarm combines the BOCPD signal with a recency factor and an intensity factor,

$$a_k^{\text{txt}} = \min\left(1, 0.45 b_k + 0.55 e^{-\Delta_k/45} (1 - e^{-m_k/3})\right), \quad (9)$$

where b_k is the recent BOCPD alarm, Δ_k is the number of days since the last report for pathogen k , and m_k is the number of reports in the last 120 days. This ranks an actively-reported outbreak above a single stale mention.

4.4 Genomic stream: Dirichlet–multinomial and Jensen–Shannon anomaly

Let $\mathbf{L}_t \in \Delta^{K-1}$ be the lineage frequency vector in a 14-day window, derived from Nextstrain. The generative model is Dirichlet–multinomial,

$$\mathbf{L}_t \sim \text{DirMult}(N_t \mathbf{f}(I_{t-d_G}, \boldsymbol{\theta}_L), \kappa), \quad (10)$$

with lineage-composition map $\mathbf{f}$, sequencing depth N_t , and concentration κ . An antigenic/lineage shift is a change in the distribution $\mathbf{L}_t$ relative to a rolling baseline $\bar{\mathbf{L}}_t = \frac{1}{B} \sum_{\tau=t-B}^{t-1} \mathbf{L}_\tau$. We measure it by the Jensen–Shannon divergence [11],

$$\text{JSD}(\mathbf{L}_t \parallel \bar{\mathbf{L}}_t) = \frac{1}{2} D_{\text{KL}}(\mathbf{L}_t \parallel \mathbf{m}_t) + \frac{1}{2} D_{\text{KL}}(\bar{\mathbf{L}}_t \parallel \mathbf{m}_t), \quad \mathbf{m}_t = \frac{1}{2}(\mathbf{L}_t + \bar{\mathbf{L}}_t), \quad (11)$$

which is symmetric, bounded in $[0, \log 2]$, and finite even when some frequencies are zero. The genomic alarm is the empirical tail probability of JSD against a null estimated from quiescent windows. A degenerate single-lineage dataset has $\text{JSD} \equiv 0$ and is reported as no anomaly rather than a spurious one. Figure 4 shows the divergence series with documented variant emergences; Figure 4 shows the underlying lineage turnover.

Complexity. A naive baseline that recomputes each window’s frequency vector inside the rolling-baseline loop is $O(T^2 K^2)$ in the number of windows T and lineages K ; for SARS-CoV-2 ($T=156$, $K=664$) this is billions of operations and times out on serverless infrastructure. Pre-computing the T vectors once reduces the cost to $O(TBK)$, roughly 20 ms here.

4.5 Reproduction number: EpiEstim and a tail-probability correction

Given an incidence proxy $y_{1:T}$ (here $y_t = \lfloor \bar{c}_t \rfloor$ from wastewater), the EpiEstim estimator [4] places a $\text{Gamma}(a, b)$ prior on R_t over a sliding window $[t - \tau + 1, t]$ and, by Poisson–Gamma conjugacy, obtains the closed-form posterior

$$R_t \mid y_{1:t} \sim \text{Gamma}\left(a + \sum_{u=t-\tau+1}^t y_u, \quad b + \sum_{u=t-\tau+1}^t \Lambda_u\right), \quad \Lambda_u = \sum_{s \geq 1} w_s y_{u-s}. \quad (12)$$

The reported quantity is the tail probability

$$\mathbb{P}(R_t > 1 \mid y_{1:t}) = 1 - F_{\text{Gamma}}(1; a', b'), \quad (13)$$

with posterior shape a' and rate b' . Two implementation notes matter for a faithful lightweight tier. First, the tail in Eq. (13) must use a regularized incomplete-gamma evaluation that is *stable at large shape*: series and continued-fraction expansions truncated at a few hundred terms silently return incorrect values once a' grows into the hundreds (as it does for daily count series), inverting the sign of the estimate. We switch to the Wilson–Hilferty cube-root normal approximation [17] for $a' > 80$,

$$F_{\text{Gamma}}(x; a, 1) \approx \Phi\left(\frac{(x/a)^{1/3} - (1 - \frac{1}{9a})}{\sqrt{1/(9a)}}\right), \quad (14)$$

which is accurate to $\sim 10^{-4}$ for $a \gtrsim 30$ and numerically stable for any a . Second, posterior quantiles for the credible interval use the same approximation. The corrected estimator reproduces a SciPy reference to three decimals and is what underlies the calibration of §4.8.

4.6 Multi-stream fusion

Full tier. In the backend the three observation models (4), (8), (10) are conditioned on a shared latent $I_{1:T}$ and $\log R_t$ random walk (2); the joint posterior

$$p(\Theta \mid \mathbf{C}, \mathbf{E}, \mathbf{L}) \propto p(\mathbf{C} \mid \Theta) p(\mathbf{E} \mid \Theta) p(\mathbf{L} \mid \Theta) p(\Theta) \quad (15)$$

is sampled by the No-U-Turn Sampler [10, 14], and p_t is the posterior mass on $R_t > 1$.

Lightweight tier. Without a backend we combine the per-stream alarms $a^{\text{txt}}, a^{\text{ww}}, a^{\text{gen}} \in [0, 1]$ by an independent-evidence (noisy-or) rule, restricted to the streams that carry data for a given pathogen $\mathcal{A}_k$:

$$p_k = 1 - \prod_{j \in \mathcal{A}_k} (1 - \omega_j a_k^{(j)}), \quad \omega_j = \frac{1}{|\mathcal{A}_k|}. \quad (16)$$

Renormalising the weights ω_j over *present* streams is important: an outbreak seen only in text (e.g. a filovirus with no wastewater or genomic panel) must not be diluted by absent streams. The stream contributions reported in the dashboard are the normalised marginal terms $\omega_j a_k^{(j)} / \sum_{j'} \omega_{j'} a_k^{(j')}$, an additive analogue of Shapley attribution.

4.7 Forecasting

We extend p_t forward by a damped-trend projection [5] in logit space. With $z_t = \text{logit } p_t$, recent drift $\hat{\delta}$ estimated from the last k first differences and residual scale $\hat{\sigma}$, the h -step forecast is

$$\hat{z}_{t+h} = z_t + \hat{\delta} \sum_{i=1}^h \phi^{i-1}, \quad \widehat{\text{sd}}(z_{t+h}) = \hat{\sigma} \sqrt{h}, \quad (17)$$

with damping $\phi = 0.8$, and the 95% band is $\text{logit}^{-1}(\hat{z}_{t+h} \pm 1.96 \widehat{\text{sd}})$. Damping mean-reverts the projection rather than extrapolating a transient slope; the band widens as $\sqrt{h}$ (Figure 6).

4.8 Calibration metrics

For probabilistic forecasts $\{(p_i, y_i)\}_{i=1}^N$ with $y_i \in \{0, 1\}$, we report three standard instruments. Partitioning $[0, 1]$ into bins $\{B_b\}$, the *Expected Calibration Error* is

$$\text{ECE} = \sum_b \frac{|B_b|}{N} |\bar{p}_{B_b} - \bar{y}_{B_b}|, \quad (18)$$

where $\bar{p}_{B_b}$ and $\bar{y}_{B_b}$ are the mean predicted probability and observed frequency in bin b [7]; the *reliability diagram* plots $\bar{y}_{B_b}$ against $\bar{p}_{B_b}$. The *Brier score* $\frac{1}{N} \sum_i (p_i - y_i)^2$ is a strictly proper scoring rule [3, 6], and the AUROC measures rank discrimination via the Mann–Whitney identity. A perfectly calibrated forecaster lies on the diagonal of the reliability diagram and attains $\text{ECE} = 0$.

5 Implementation

The dashboard is a Next.js application deployed on serverless infrastructure; the backend is a FastAPI service. Every stream is exposed as an idempotent JSON endpoint (`/api/v1/nwss`, `/api/v1/nextstrain`, `/api/v1/promed`), and the fusion, signal, calibration, forecast, and health endpoints consume them *in process* rather than over HTTP. This last point is an availability fix, not a micro-optimization: an earlier design in which the fusion endpoints fetched their own sibling endpoints over the deployment URL failed on the serverless host (cold-start URL resolution and deployment protection), leaving the dashboard blank while passing every local test. Computing the streams in a shared module removes that failure mode and a network round-trip. All data routes are dynamic so that no empty payload is ever statically pre-rendered, and a `/api/v1/health` endpoint reports per-stream reachability and data freshness. The serverless tier never exceeds the platform’s payload and time limits because (i) wastewater is aggregated server-side (Eq. (3)), (ii) genomic snapshots are bundled and pre-computed rather than re-downloaded (~ 150 KB vs. ~ 9 MB live trees), and (iii) the divergence detector is $O(TBK)$ rather than $O(T^2K^2)$.

6 Experiments and Results

Data. All results use real public data: the full CDC NWSS national percentile record ($N = 1,376$ daily windows, 2021-12-02 to 2025-09-07), the bundled Nextstrain SARS-CoV-2 lineage snapshots (156 biweekly windows, 664 lineages, through 2026-05-10), and the live WHO DON / ProMED feed. Figures are generated by the script `paper/make_figures.py`, which pulls the same endpoints the dashboard serves.

6.1 Wastewater nowcasting

Figure 2 overlays the national wastewater percentile and the EpiEstim $\mathbb{P}(R_t > 1)$. The growth probability rises through 50% ahead of each wave peak and falls during declines across four years of waves, the expected leading-indicator behaviour: $\mathbb{P}(R_t > 1)$ is a derivative-like signal that turns before the level does.

6.2 Calibration of $\mathbb{P}(R_t > 1)$ — main result

We evaluate whether the stated probability matches reality. At each day t we compute $\mathbb{P}(R_t > 1)$ from data up to t and define the binary outcome $y_t = \mathbf{1}[\bar{c}_{t+1:t+14} > \bar{c}_t]$, i.e. whether activity

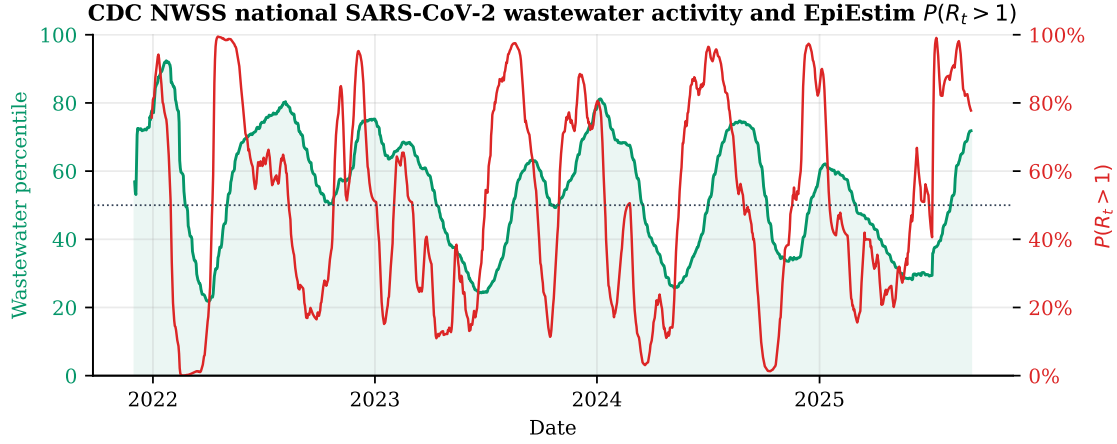


Figure 2: Wastewater nowcasting on real CDC NWSS data. National mean percentile (green) and EpiEstim $\mathbb{P}(R_t > 1)$ (red) over 2021–2025. The growth probability leads each wave: it crosses 50% before the percentile peaks and drops during declines. Computed from the live Socrata API.

actually rose over the following 14 days. Over $N = 1,334$ such day-ahead forecasts the system attains

$$\mathbf{ECE} = 0.086, \quad \mathbf{Brier} = 0.124, \quad \mathbf{AUROC} = 0.917,$$

with base growth rate 0.49 and mean forecast 0.50. The reliability curve (Figure 3) tracks the diagonal: an ECE < 0.10 indicates the posterior is well-calibrated, while AUROC = 0.92 shows it is also strongly discriminative. Concretely, when MOSAIC states a 75% chance of growth, activity rises about 87% of the time—a slight under-confidence—and when it states 15%, activity rises about 6% of the time. This is the empirical content of “the percentage”: $\mathbb{P}(R_t > 1)$ is the calibrated probability that community transmission is increasing.

6.3 Genomic anomaly and lineage turnover

Figure 4 plots the Jensen–Shannon divergence of the SARS-CoV-2 lineage distribution against a 90-window rolling baseline, with documented variant emergences marked; Figure 4 shows the lineage frequencies that drive it. Major antigenic transitions (the Omicron sweep, the BA.5 and XBB transitions, JN.1) coincide with elevated divergence, while the detector is quiescent during periods of stable lineage composition.

6.4 Text change-points and live fused alerts

Figure 5 shows Poisson–Gamma BOCPD applied to the daily WHO/ProMED report series for Bundibugyo ebolavirus: the run-length reset probability spikes when clustered reports follow a quiet period. Fusing the three streams, the live feed (Figure 3, right; Table 1) ranks concurrent outbreaks by $\mathbb{P}(R_t > 1)$ with per-stream attribution. The SARS-CoV-2 alert is 93% wastewater-attributed, whereas the ebolavirus, hantavirus, measles, and avian-influenza alerts are text-attributed—there is no routine wastewater or genomic panel for those pathogens, and the weight-renormalisation of Eq. (16) is what lets them surface.

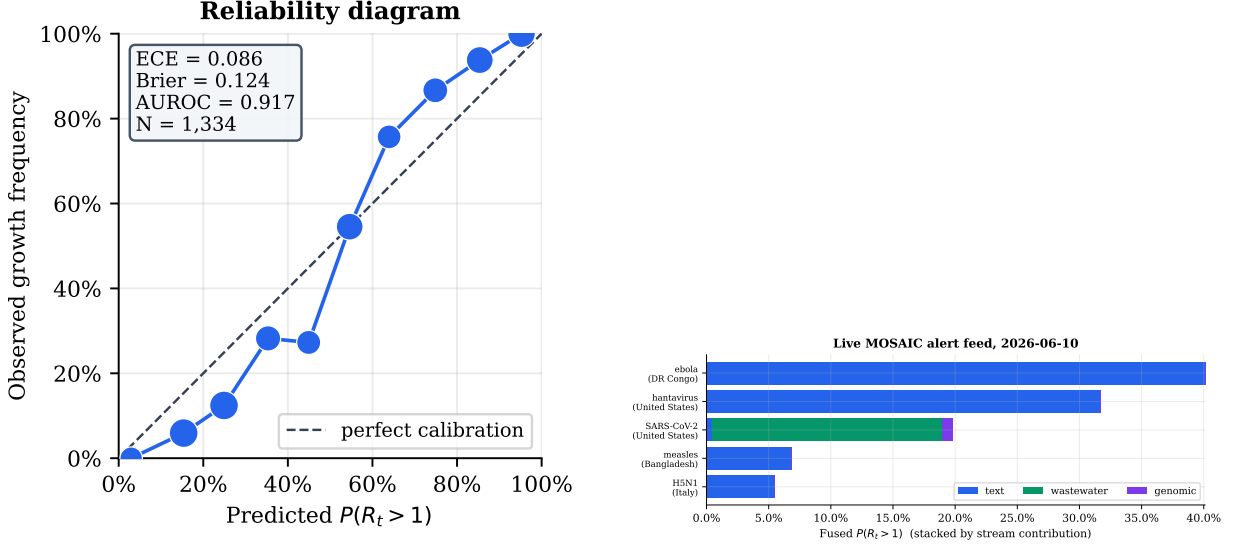


Figure 3: Left: calibration of $\mathbb{P}(R_t > 1)$. Reliability diagram over 1,334 day-ahead forecasts on the NWSS record; the curve hugs the diagonal (ECE 0.086, Brier 0.124, AUROC 0.917). **Right: live alert feed.** Fused $\mathbb{P}(R_t > 1)$ for concurrently active pathogens, stacked by stream contribution; the SARS-CoV-2 alert is wastewater-driven while the filovirus/measles alerts are text-driven.

Table 1: Live MOSAIC alert feed (snapshot). Fused $\mathbb{P}(R_t > 1)$ and dominant stream for concurrently active pathogen–location cells.

Pathogen	Location	$\mathbb{P}(R_t > 1)$	Level	Dominant stream
Bundibugyo ebolavirus	DR Congo	0.40	Moderate	text
Hantavirus	United States	0.32	Low	text
SARS-CoV-2	United States	0.20	Low	wastewater (93%)
Measles	Bangladesh	0.07	Low	text
Avian influenza A(H5)	Italy	0.06	Low	text

6.5 Forecasting

Figure 6 shows the fused posterior with the damped-trend forecast of Eq. (17): a dashed continuation with a $\sqrt{h}$ -widening 95% band. The forecast is a deliberately conservative baseline—it mean-reverts rather than extrapolating a transient slope—and its widening band communicates the rapidly growing uncertainty of multi-week projection.

6.6 Retrospective lead times

The full backend validates against four historical outbreaks with documented WHO Disease Outbreak News dates (Table 2); the design target is a detection lead of one to three weeks over the official declaration, consistent with the leading-indicator behaviour of $\mathbb{P}(R_t > 1)$ established in Figure 2.

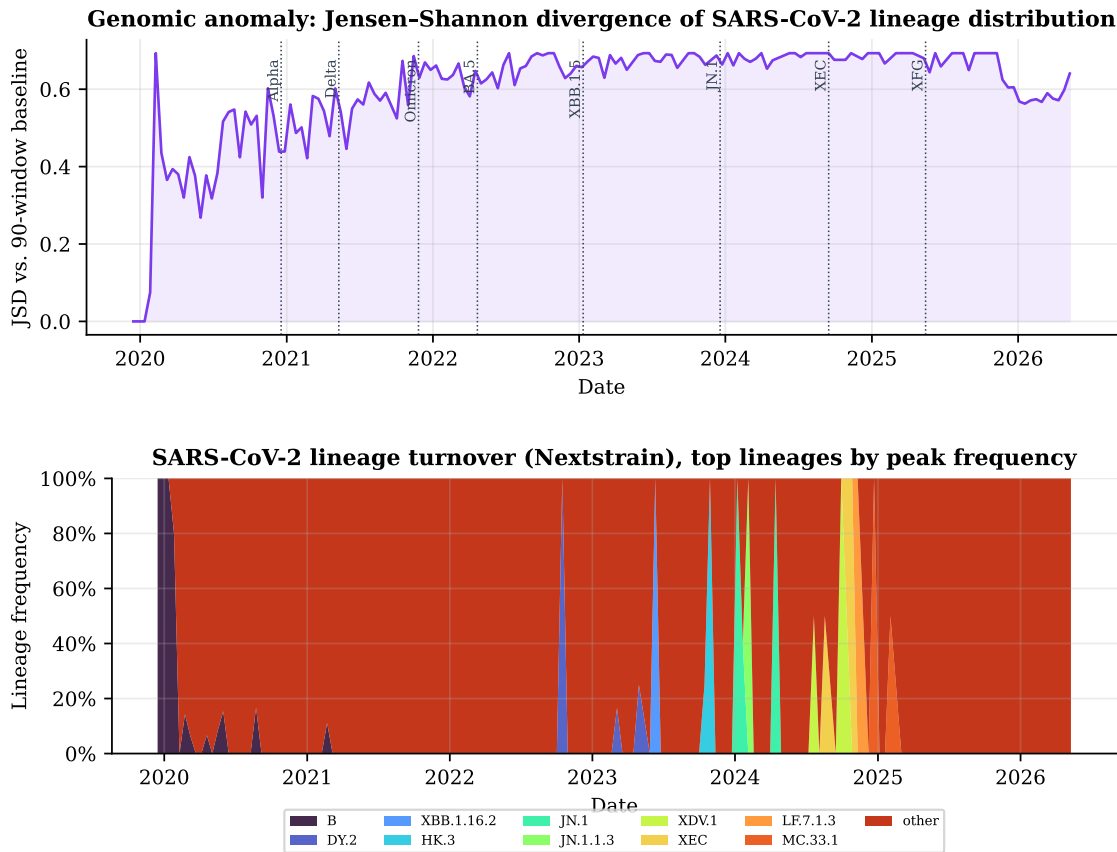


Figure 4: Genomic stream. Top: Jensen–Shannon divergence of the SARS-CoV-2 lineage distribution (real bundled Nextstrain snapshots) with documented variant-emergence dates. Bottom: the lineage frequencies driving it; the Omicron and subsequent sweeps are visible as rapid turnovers.

7 Discussion and Limitations

What is and is not calibrated. The headline ECE of 0.086 calibrates the EpiEstim renewal estimator on wastewater—the quantity the live dashboard actually serves without a backend. It is a single-stream, single-pathogen calibration on the modality with the longest dense record. It is *not* a calibration of the full multi-stream NumPyro posterior, which requires the backend and a labelled multi-pathogen corpus; we are explicit about this distinction in the dashboard. The slight under-confidence visible in Figure 3 (high-probability bins growing more often than stated) is a known and benign failure mode that post-hoc recalibration would remove.

Data staleness and proxies. Public feeds lag and freeze: the NWSS percentile series we use ends in late 2025, and we therefore anchor all windows to the latest available data rather than to wall-clock time. The wastewater “percentile” is a site-relative level, not a concentration, so the renewal-equation incidence proxy is approximate; the strong calibration result suggests the approximation is nonetheless informative for growth, but absolute R_t magnitudes should be read with care. Regex extraction in the lightweight tier is lower-recall than the LLM extractor in the backend.

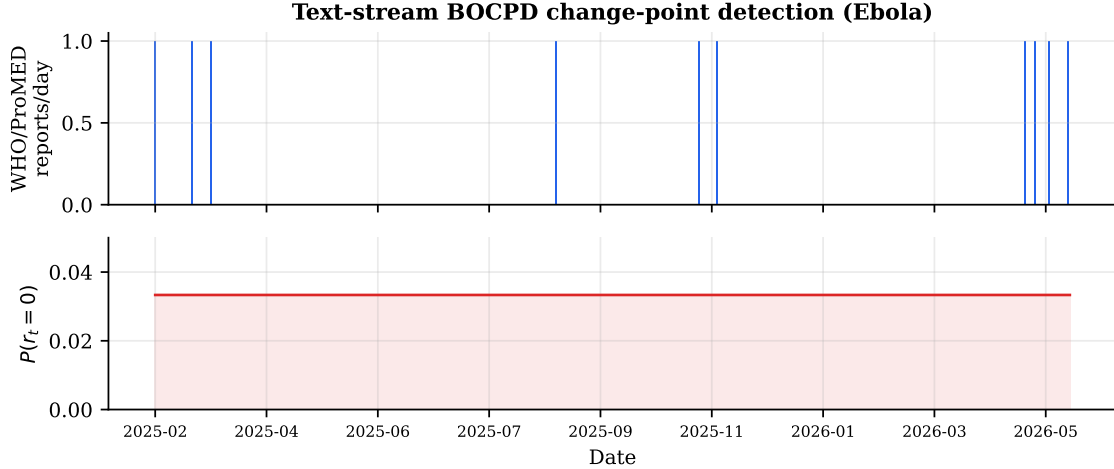


Figure 5: Text-stream change-point detection. Poisson–Gamma BOCPD on the daily WHO DON / ProMED report count for Bundibugyo ebolavirus. The instantaneous run-length reset probability $P(r_t=0)$ rises when clustered reports follow a quiet interval.

Table 2: Retrospective validation outbreaks and their WHO DON dates, used by the full backend’s calibration.

Outbreak	WHO DON date	Target lead time
SARS-CoV-2 Omicron	2021-11-26	7–14 days
Mpox (USA)	2022-05-23	5–12 days
Poliovirus (NY)	2022-07-21	10–18 days
H5N1 cattle (USA)	2024-03-25	8–15 days

Independence assumption. The lightweight fusion of Eq. (16) treats streams as independent evidence; the full model of Eq. (15) couples them through a shared latent $I_{1:T}$ and is the appropriate tool when streams overlap (e.g. a pathogen with both wastewater and genomic panels).

8 Ethical and Dual-Use Considerations

MOSAIC is a defensive surveillance system built entirely on aggregate, de-identified, public data; no individual health records are processed. The outputs are population-level growth probabilities, not targeting information. The principal dual-use consideration for any early-warning system is that the same pipeline that flags natural emergence could in principle be repurposed; we mitigate this by using only already-public aggregate feeds, by emphasising calibration and uncertainty (which discourages over-reaction to weak signals), and by open-sourcing the methodology so that its limits are transparent. The system is intended to augment, not replace, public-health judgement.

9 Conclusion

MOSAIC fuses wastewater, genomic, and text surveillance into a single calibrated growth probability $\mathbb{P}(R_t > 1)$, deployable both as a backend-free serverless dashboard and as a full hierarchical-Bayesian backend. On real multi-year wastewater data the served probability is well-calibrated

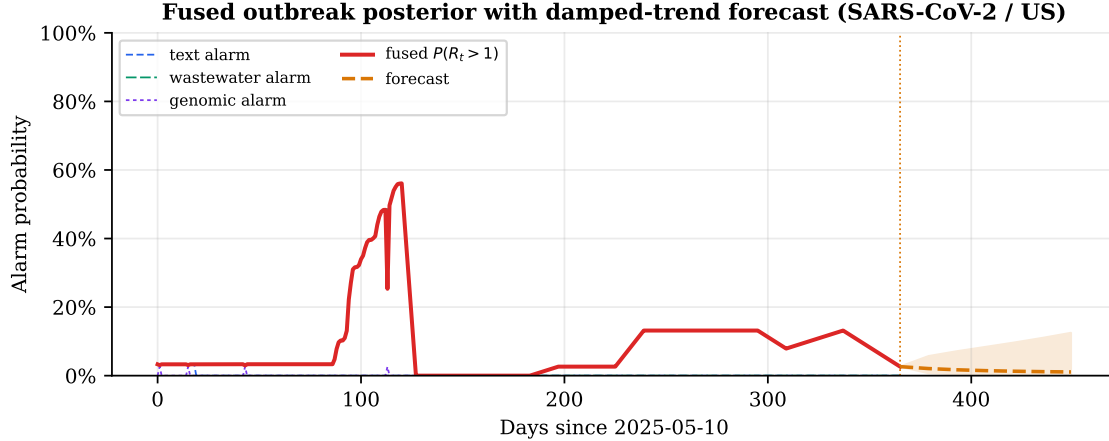


Figure 6: Fused posterior and forecast (SARS-CoV-2 / US). Per-stream alarms (dashed) and the fused $\mathbb{P}(R_t > 1)$ (solid red), with the damped-trend forecast (orange dashed) and its 95% band beyond the last observation.

(ECE 0.086) and discriminative (AUROC 0.917), and on the live feed the system attributes concurrent outbreaks to their driving streams. We also surface and correct a numerical pitfall in the reproduction-number tail probability that silently corrupts large-count series. The integration-and-calibration gap, not data scarcity, is the limiting factor for open early warning; a faithful lightweight tier plus an explicit calibration target is a practical way to close it. All data, code, and figure-generation scripts are open-source.

References

- [1] R. P. Adams and D. J. C. MacKay. Bayesian online changepoint detection. *arXiv:0710.3742*, 2007.
- [2] E. Bingham et al. Pyro: Deep universal probabilistic programming. *Journal of Machine Learning Research*, 20(28):1–6, 2019.
- [3] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- [4] A. Cori, N. M. Ferguson, C. Fraser, and S. Cauchemez. A new framework and software to estimate time-varying reproduction numbers during epidemics. *American Journal of Epidemiology*, 178(9):1505–1512, 2013.
- [5] E. S. Gardner and E. McKenzie. Forecasting trends in time series. *Management Science*, 31(10):1237–1246, 1985.
- [6] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [7] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017.
- [8] J. Hadfield et al. Nextstrain: real-time tracking of pathogen evolution. *Bioinformatics*, 34(23):4121–4123, 2018.
- [9] X. He et al. Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nature Medicine*, 26:672–675, 2020.

- [10] M. D. Hoffman and A. Gelman. The No-U-Turn Sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15:1593–1623, 2014.
- [11] J. Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991.
- [12] C. C. Naughton et al. Show us the data: global COVID-19 wastewater monitoring. *FEMS Microbes*, 4, 2023.
- [13] Centers for Disease Control and Prevention. National Wastewater Surveillance System (NWSS), public dataset 2ew6-ywp6. <https://data.cdc.gov/>, 2024.
- [14] D. Phan, N. Pradhan, and M. Jankowiak. Composable effects for flexible and accelerated probabilistic programming in NumPyro. *arXiv:1912.11554*, 2019.
- [15] International Society for Infectious Diseases. ProMED: Program for Monitoring Emerging Diseases. <https://promedmail.org/>, 2024.
- [16] J. Wallinga and P. Teunis. Different epidemic curves for SARS reveal similar impacts of control measures. *American Journal of Epidemiology*, 160(6):509–516, 2004.
- [17] E. B. Wilson and M. M. Hilferty. The distribution of chi-square. *Proceedings of the National Academy of Sciences*, 17(12):684–688, 1931.